

Notes: Igami (JPE, 2017)

Estimating the Innovator's Dilemma: Structural Analysis of Creative Destruction in the Hard Disk Drive Industry, 1981–1998

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1 Big picture

- **Question.** Why do incumbents often innovate more slowly than entrants when a new technology arrives? Is the gap driven by inferior innovative ability, by strategic incentives, or by organizational frictions?
- **Setting.** The paper studies the hard disk drive (HDD) industry and focuses on the transition from the 5.25-inch generation to the 3.5-inch generation between 1981 and 1998.
- **Core challenge.** The stylized fact is clear: only about half of incumbents ever moved into the new generation, and even those that did tended to do so later than entrants. The challenge is to separate three forces that all plausibly matter: cannibalization of old products, preemption of rivals, and differences in innovation costs between incumbents and entrants.
- **Main contribution.** The paper builds and estimates a structural dynamic oligopoly model that combines three layers: logit demand across HDD categories, Cournot competition in period profits, and a dynamic game of entry, exit, and innovation.
- **Main result.** Incumbents are not slow because they are bad innovators. The estimated model implies that incumbents actually have *lower* sunk costs of innovation than entrants, but they are still reluctant to innovate because of cannibalization.
- **Key quantitative finding.** Removing cannibalization shrinks the incumbent-entrant innovation gap by about 57%. Removing preemption widens the gap by about 38%. So preemption pushes incumbents to innovate earlier, but not enough to offset the replacement effect.
- **Key structural finding.** The baseline estimates imply a fixed operating cost of about \$0.15 billion, an incumbent innovation sunk cost of about \$1.24 billion, and an entrant entry/innovation sunk cost of about \$2.25 billion.
- **Fit.** The estimated model reproduces the broad path of market structure well: old-only firms fall quickly, both-firms stabilize around four, and new-only firms peak higher than innovative incumbents and then decline faster.
- **Policy result.** A perfectly predictable ex ante broad patent can raise welfare in the model because it avoids wasteful duplicate entry, but a more realistic ex post monopoly grant performs terribly because the sunk costs have already been incurred by the time legal exclusivity is enforced.
- **Economic takeaway.** The paper gives a rational, incentive-based explanation of the innovator's dilemma. Incumbents may innovate slowly not because of incompetence, but because the new product eats their old profits.

2 Data and measurement (key objects)

2.1 Why the HDD industry is a good laboratory

- The HDD industry experienced repeated technological transitions: 14-inch, 8-inch, 5.25-inch, 3.5-inch, and 2.5-inch generations.

- Each generation was defined by disk diameter, but commercial success still required substantial engineering work, assembly-line redesign, and reliable volume production.
- In every transition, only about half of incumbents innovated into the next generation, and even successful incumbents typically entered about two years later than entrants.
- The 5.25-inch to 3.5-inch transition is especially attractive because both generations served the same downstream market — desktop personal computers — so the paper avoids confounding differences across separate end-use segments.

Interpretation. This is a near-ideal setting for studying creative destruction because technologies and firms turn over together, product generations are clearly defined, and the timing of innovation is observable.

2.2 Data sources

The paper uses two datasets, both manually assembled from *DISK/TREND Reports*.

- **Firm panel.** A panel of the world’s HDD manufacturers, including entry, exit, and production status by generation. The author digitizes 1,378 firm-year observations overall.
- **Sales panel.** Aggregate prices and shipment quantities by product category, where a category is a generation-quality pair observed by region, user type, and year.

The main estimation sample isolates the 5.25-inch and 3.5-inch generations and spans:

- 18 years (1981–1998),
- 259 firm-years in the firm panel,
- 405 product-category observations in the sales data used for the static demand system.

Interpretation. The data are unusually rich on market structure and timing but limited on within-generation firm heterogeneity because sales are observed at the category level, not at the firm-brand level.

2.3 How innovation and firm types are defined

A firm is classified into one of four technological states:

- **old only:** incumbent producing only 5.25-inch HDDs,
- **both:** incumbent that has innovated and now produces both 5.25-inch and 3.5-inch HDDs,
- **new only:** entrant producing only 3.5-inch HDDs,
- **potential entrant:** a firm that has announced intent to enter but has not yet begun shipping.

The innovation event is the first shipment of 3.5-inch HDDs. Potential entrants are identified from industry reports that record public announcements of intended product specifications before actual production.

Interpretation. The paper separates *innovating incumbents* from *entering new firms*. That distinction is central because the whole question is why incumbents are slower than entrants.

2.4 Descriptive evidence on the innovation gap

Table 1 summarizes who eventually moved into the 3.5-inch generation.

- Of 66 firms that entered initially in 5.25-inch HDDs, only 19 ever produced 3.5-inch HDDs, a fraction of 28.8%.
- Of 41 firms that entered initially in 14-inch HDDs, only 10 eventually moved to 3.5-inch HDDs.
- By contrast, among firms that entered the HDD market directly through the 3.5-inch generation, 31 of 36 are observed producing 3.5-inch drives.
- Asian firms look somewhat more likely to adopt the new generation than U.S. firms, but most other background characteristics do not strongly predict timing once the paper moves to its structural setup.

Figure 1 makes the central stylized fact visual.

- By 1998, about 12 entrants had innovated into 3.5-inch HDDs, but only 8 incumbents had done so.
- Measured as a fraction of all firms that *could* have innovated, the entrant path is still far above the incumbent path.

Interpretation. The paper's target is not merely that entrants exist. It is that incumbents leave a lot of innovation "on the table" even though they already know the industry.

2.5 Aggregate market data and key trends

Table 2 reports the main aggregate sales objects.

- The average transaction price is around \$882 in 1998 dollars.
- The average storage capacity is about 2,122 megabytes across category-year observations, with very large dispersion as the technology frontier moves.
- Half of the observations correspond to the 3.5-inch generation.

Figure 3 shows the main industry trends:

- shipment quantities rise quickly,
- prices trend downward,
- storage capacity improves steadily,
- market structure rotates from old-only firms toward both-firms and then toward new-only firms.

Interpretation. These trends motivate the paper's main simplifying assumptions: rapid exogenous demand growth, falling unit costs, and a nonstationary environment in which a finite-horizon dynamic game is more natural than a stationary infinite-horizon model.

3 Models and empirical specifications (all equations + interpretation)

3.1 Timing, states, and order of moves

Time is discrete with a finite horizon $t = 0, 1, 2, \dots, T$. The industry state is summarized by the counts

$$s_t = \left(N_t^{old}, N_t^{both}, N_t^{new}, N_t^{pe} \right), \quad (1)$$

where the four components are the numbers of old-only firms, both-firms, new-only firms, and potential entrants.

Within a year:

- firms first earn static period profits from product-market competition,
- then old-only incumbents choose among {exit, stay, innovate},
- then both-firms choose among {exit, stay},
- then new-only firms choose among {exit, stay},
- then potential entrants choose among {quit, enter}.

Each active or potential firm receives iid extreme-value private cost shocks tied to the discrete options.

Interpretation. The order-of-moves assumption keeps the dynamic game computationally feasible because each type effectively solves a single-agent problem conditional on expectations about later movers.

3.2 Static period profits

Static profits depend on market structure, demand, and costs:

$$\pi_{it} = \pi_t(s_{it}, s_{-it}; D_t, C_t). \quad (2)$$

The economic meaning is straightforward:

- demand determines the prices and substitution patterns between old and new HDDs,
- costs determine the profitability of supplying each generation,
- the number of rivals determines how much profit any active firm can earn.

Interpretation. Cannibalization enters here through substitution between old and new HDDs. Pre-emption enters because more firms in the new generation lower future profits.

3.3 Demand system

The paper estimates a multinomial logit demand system over HDD product categories. Buyer k 's utility from category j is

$$u_{kj} = a_0 + a_1 p_j + a_2 I(g_j = \text{new}) + a_3 x_j + \xi_j + \varepsilon_{kj}, \quad (3)$$

where:

- p_j is price,
- $I(g_j = \text{new})$ is an indicator for the 3.5-inch generation,
- x_j is log storage capacity,
- ξ_j captures unobserved quality,
- and the outside good includes removable HDDs and other storage devices.

Berry inversion gives the estimating equation

$$\ln \left(\frac{ms_j}{ms_0} \right) = a_1 p_j + a_2 I(g_j = \text{new}) + a_3 x_j + \xi_j, \quad (4)$$

where ms_j is the market share of category j and ms_0 is the outside-good share.

The preferred IV-logit estimate under the narrow market definition gives:

- price coefficient -3.28 ,
- 3.5-inch dummy coefficient 0.91 ,
- log-capacity coefficient 1.20 .

Interpretation. Demand slopes down in price, buyers like higher capacity, and the 3.5-inch generation carries an independent utility premium after controlling for price and capacity. The substitution estimates are what translate incumbent innovation into cannibalization risk.

3.4 Cournot competition and marginal costs

Within each generation, the paper assumes homogeneous Cournot competition. Firm i 's profit is

$$\pi_i = \sum_{g \in A_i} (p_g - mc_g) q_{ig}, \quad (5)$$

where A_i is the set of generations produced by firm i .

For a multiproduct firm, the first-order condition is

$$p_g + \frac{\partial p_g}{\partial Q_g} q_{ig} + \frac{\partial p_h}{\partial Q_g} q_{ih} = mc_g, \quad (6)$$

with $g, h \in \{\text{old}, \text{new}\}$ and $g \neq h$.

Using the demand estimates and these FOCs, the paper backs out annual marginal costs mc_t^{old} and mc_t^{new} .

Interpretation. This step is crucial because the dynamic game is estimated from choice data, but the payoffs that rationalize those choices come from the static demand-and-cost system.

3.5 Dynamic optimization

Old-only incumbents solve

$$V_t^{old}(s_t, \varepsilon_{it}) = \pi_t^{old}(s_t) + \max \left\{ \begin{aligned} &\varepsilon_{it}^0, \\ &-f + \beta E[V_{t+1}^{old}(s_{t+1}, \varepsilon_{i,t+1}) \mid s_t, \varepsilon_{it}] + \varepsilon_{it}^1, \\ &-f + \beta E[V_{t+1}^{both}(s_{t+1}, \varepsilon_{i,t+1}) \mid s_t, \varepsilon_{it}] - k^{inc} + \varepsilon_{it}^2 \end{aligned} \right\}. \quad (7)$$

Post-innovation incumbents and new-only entrants choose between exit and stay, while a potential entrant solves

$$\max \left\{ \varepsilon_{it}^0, \beta E[V_{t+1}^{new}(s_{t+1}, \varepsilon_{i,t+1}) \mid s_t, \varepsilon_{it}] - k^{ent} + \varepsilon_{it}^1 \right\}. \quad (8)$$

The dynamic parameters are:

- f : fixed cost of continued operation,
- k^{inc} : sunk innovation cost for incumbents,
- k^{ent} : sunk entry/innovation cost for potential entrants.

Interpretation. Cannibalization appears because becoming a both-firm changes future own profits. Preemption appears because innovating now changes rivals' future incentives to enter or innovate.

3.6 Likelihood-based structural estimation

Because the private cost shocks are extreme value, the model yields closed-form conditional choice probabilities. The likelihood over observed market structures, exits, and innovations is

$$P(N, X, E) = \prod_{t=0}^{T-1} P(N_t, X_t, E_t), \quad (9)$$

and the estimator is

$$\arg \max_{f, k^{inc}, k^{ent}} \ln P(N, X, E). \quad (10)$$

The paper does *not* estimate the discount factor β because identification is weak in this class of models. Instead it calibrates β , the time trend in innovation costs δ , and the number of potential entrants N^{pe} , then studies sensitivity.

Interpretation. The dynamic estimation is not trying to identify every primitive from scratch. It uses a disciplined mix of static estimation, likelihood-based dynamic estimation, and calibrated parameters where identification is known to be weak.

4 Identification and instruments (what makes the estimates credible)

4.1 What identifies the demand estimates

The demand system uses three kinds of variation:

- time-series changes in prices and category shares,
- cross-sectional variation across generation-quality bins,
- cross-sectional variation across region and user type under the narrow market definition.

The main price instruments are:

- prices in the other region/user market (Hausman-Nevo type instruments),
- the number of product models in a category,
- and, in a robustness exercise, time-series orthogonality restrictions based on an AR(1) process for unobserved quality.

Interpretation. The identifying idea is standard differentiated-products logic: cost shocks can be correlated across markets, while taste shocks are less likely to be.

4.2 What identifies marginal costs and period profits

Once demand is estimated, marginal costs are backed out from Cournot first-order conditions under three important assumptions:

- products are commodity-like within generation,
- firms are symmetric within type,
- marginal cost is common across firms and constant in quantity within a year-generation.

Interpretation. The paper deliberately gives up rich within-generation heterogeneity because the data do not report firm-level shares by product category. That simplification is the price paid for structural tractability.

4.3 What identifies the dynamic parameters

The dynamic parameters are pinned down by how often firms in each state choose exit, stay, adopt, or enter.

- More exit among active firms pushes up the estimate of f .
- Fewer incumbent innovations push up k^{inc} , holding period profits fixed.
- Fewer entrant innovations push up k^{ent} , holding period profits fixed.

Interpretation. The logic is clean: static profits tell us what firms *stand to gain*; observed dynamic choices tell us how large the missing fixed and sunk costs must be to rationalize what they actually do.

4.4 Why reverse causality is less problematic here

The paper's core object is not a reduced-form regression of innovation on market outcomes. Instead, the direction of causality is disciplined by the model structure:

- product-market outcomes determine period profits,
- period profits feed into dynamic values,
- dynamic values generate entry, exit, and innovation probabilities.

There is still model risk, of course, but the key point is that the paper is not relying on a naive contemporaneous correlation between innovation and profits.

4.5 Why the paper emphasizes rational incentives rather than managerial biases

A major contribution of the paper is to show that the innovator's dilemma can arise even under rational expectations and without assuming that incumbents are cognitively biased or organizationally incompetent.

At the same time, the author is careful about what the estimates can and cannot distinguish:

- if cognitive or organizational biases were important in the data, they would be absorbed into the estimated sunk costs;
- therefore $k^{inc} < k^{ent}$ should be read as a net innovation-cost comparison after all such forces are bundled together.

Interpretation. This is why the result is so strong. Even after allowing the residual cost terms to absorb many unmodeled disadvantages, incumbents still look like the cheaper innovators.

4.6 Alternative explanations and limits

The paper explicitly discusses two alternative families of explanations.

- **Christensen-style cognitive and organizational bias.** The paper argues these may exist, but they are not needed to explain the main pattern.
- **Jovanovic-style learning and selection.** That approach can explain why older incumbents may be larger or more productive, but it has trouble explaining the sharp role of new competition and the technology transition itself.

Important limits remain:

- the model does not include endogenous mergers,
- it does not model intensive-margin R&D investment,
- it abstracts from persistent firm heterogeneity within type,
- and the patent-policy experiments are stylized thought experiments rather than literal policy evaluations.

4.7 Relation to the literature

The paper sits at the intersection of three literatures:

- **creative destruction and innovation:** Arrow, Schumpeter, Christensen,
- **dynamic oligopoly and entry/exit:** Benkard, Schmidt-Dengler, Goettler and Gordon, Seim, Aguirregabiria and Mira,
- **industry dynamics and technology diffusion:** Jovanovic, Hopenhayn, Klepper, and related work.

Interpretation. The paper's distinctive move is to put these pieces together in one structural setting where the competition between incumbents and entrants is front and center.

5 Tables and figures

The exhibits line up tightly with the paper's narrative: first document the innovation gap, then motivate the HDD transition as a clean empirical laboratory, then estimate static profits, then recover dynamic costs, and finally use counterfactuals to separate cannibalization, preemption, and policy effects.

5.1 Figure 1: the innovation gap is the fact to be explained

Read:

- The top panel plots cumulative innovators. Entrants reach 12 innovators by the end of the sample, while incumbents flatten out at 8.
- The bottom panel normalizes by the relevant pools of firms and tells the same story: entrants innovate much more aggressively even after correcting for the different number of potential innovators.

Economic message. Figure 1 is the whole empirical puzzle in one picture. The rest of the paper asks why incumbents innovate less even though they know the industry and, as the later estimates suggest, are not less efficient at innovating.

5.2 Figure 2, Table 1, Table 2, and Figure 3: why this industry is a canonical case of creative destruction

Read:

- Figure 2 shows the succession of HDD generations. The 5.25-inch generation rises quickly, peaks, and then declines as 3.5-inch drives become dominant.
- Table 1 shows that fewer than 30% of firms entering through the 14-inch, 8-inch, or 5.25-inch generations ever made it into 3.5-inch HDDs.
- Table 2 shows that the sales data contain substantial variation in prices, quantities, and storage capacity across category-year observations.
- Figure 3 shows the key market trends that motivate the model: quantities surge, prices fall, quality rises roughly monotonically, and market structure shifts from old-only firms to both-firms and then to new-only firms.

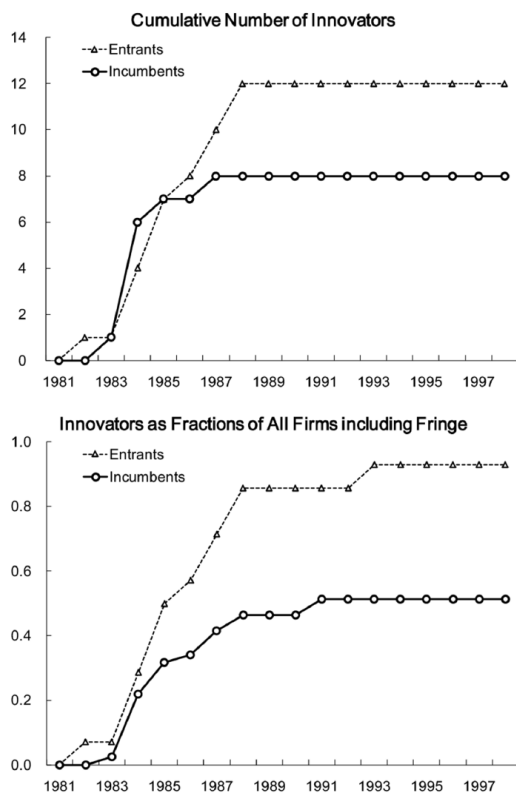


FIG. 1.—The incumbent-entrant innovation gap. The top panel plots the timing of the initial shipment of 3.5-inch HDDs separately for incumbents (i.e., firms already active in the 5.25-inch generation) and entrants (i.e., firms that appeared for the first time as producers of 3.5-inch HDDs). The bottom panel expresses similar numbers in terms of the

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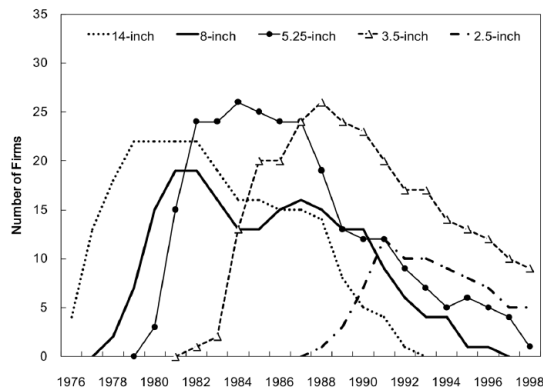


FIG. 2.—Shifting generations of technology. Multigeneration firms are counted multiple times. Mergers and acquisitions were not a major channel of exit during the sample period, but their importance grew in the 2000s. See Igami and Uetake (2015) for a dynamic structural analysis of endogenous mergers and innovation in this industry.

namic part of my model. Third, I embed these implied period profits into a dynamic oligopoly game of entry, exit, and innovation and estimate its key parameters (the firms' fixed costs of operation, as well as the incum-

FIRM CHARACTERISTICS	NUMBER OF FIRMS	EVER PRODUCED 3.5-INCH		INITIAL 3.5-INCH PRODUCTION	
		COUNT	Fraction (%)	Mean Year	Standard Deviation
Initial diameter of entry:					
14-inch	41	10	24.4	1985.5	2.3
8-inch	21	3	14.3	1987.5	3.2
5.25-inch	66	19	28.8	1986.6	2.5
3.5-inch	36	31	86.1*	1987.3	3.3
Other	14	0	0	—	—
Organizational form:					
Specialized HDD start-up	74	24	32.4	1986.9	3.5
Computer maker	52	19	36.5	1986.6	2.3
HDD component maker	13	2	15.4	1991.0	5.7
Other electronics maker	39	17	43.4	1986.4	2.3
Region of origin:					
United States	101	22	21.8	1986.2	2.6
Asia	46	30	65.2	1987.1	2.9
Europe (west)	18	4	22.2	1986.8	4.9
Europe (east)	3	0	0	—	—
Brazil	10	6	60.0	1987.0	3.3
Total	178	62	34.8	1986.8	2.9

SOURCE.—DISK/TREND Reports.
* Less than 100% of the firms that announced intentions to produce 3.5-inch HDDs actually did.

Variable	Observations	Standard			
		Mean	Deviation	Minimum	Maximum
Shipment (million units)	405	1,405	5,339	.1	28,332
Average price (1998 constant US\$)	405	892	892	72	7,487
Average quality (information storage capacity in megabytes)	405	2,122	5,922	15	60,000
Diameter 3.5 inches	405	.49	.50	0	1

SOURCE.—DISK/TREND Reports.
NOTE.—Unit of observation is product category (diameter-quality pairs), aggregated across firms and models, by year and segment.

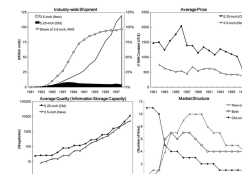


FIG. 3.—Quality, price, quality and volume convergence. Both 5.25- and 3.5-inch HDDs were in the same market, namely, desktop PCs. Quality is measured by average capacity per unit for each generation. "On-line" and "server" firm produce 3.5- and 5-inch HDDs, respectively. "Bolt" represents innovations that adapted for use in desktop, server producing both of the two generations. "Personal server" is identified by the announcement of product specification (without actual shipment).

Economic message. These exhibits do two jobs. They show that the transition is economically major, and they justify the paper’s modeling choice to focus on the 5.25-inch versus 3.5-inch contest in desktop PCs rather than on all HDD generations at once.

5.3 Table 3 and Figure 4: the static model says the new generation is attractive, but not for free

TABLE 3
LOGIT DEMAND ESTIMATES FOR 5.25- AND 3.5-INCH HDDs

	BROAD MARKET		NARROW MARKET	
	OLS (1)	IV (2)	OLS (3)	IV (4)
Price (\$000)	-1.66*** (.45)	-2.99*** (.55)	-.93** (.46)	-3.28*** (.63)
Diameter 3.5 inches	.84* (.46)	.75 (.45)	1.75*** (.31)	.91** (.38)
Log capacity (megabytes)	.18 (.33)	.87*** (.27)	.04 (.26)	1.20*** (.31)
Year dummies	Yes	Yes	Yes	Yes
Region/user dummies	...	...	Yes	Yes
Adjusted R^2	.43	.33	.50	.28
Observations	176	176	405	405
Partial R^2 for price	...	.32	...	.16
p -value	...	.00	...	.00

NOTE.—Standard errors, in parentheses, are clustered by capacity-diameter (cols. 1 and 2) and capacity-diameter-region-user (cols. 3 and 4).

* Significant at the 10 percent level.

** Significant at the 5 percent level.

*** Significant at the 1 percent level.

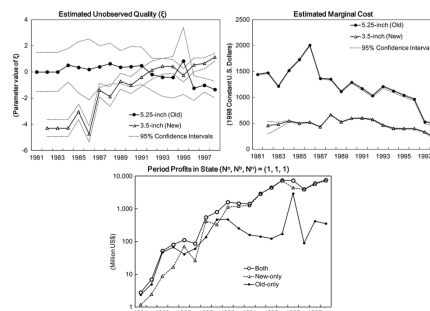


FIG. 4.—Estimated unobserved quality, marginal costs, and period profits. Results based on the IV estimates of logit demand system with the narrow market definition (col. 4 of table 3).

Read:

- Table 3 gives the preferred IV-logit demand estimates. The price coefficient is strongly negative, while both the 3.5-inch indicator and log capacity are positive and significant in the preferred specification.
- Figure 4 then translates these estimates into recovered unobserved quality, marginal costs, and implied period profits.
- The left panel of Figure 4 shows that the unobserved quality ranking of 3.5-inch versus 5.25-inch HDDs flips over time, with the 3.5-inch generation becoming the mainstream technology.
- The right panel shows declining marginal costs for both generations over time.
- The bottom panel shows that a both-firm earns more period profit than an old-only firm in every year.

Economic message. This is an important milestone in the paper. Static profits do *not* say incumbents should avoid innovation. The reluctance only appears once dynamic cannibalization and strategic interactions are introduced.

5.4 Table 4, Table 5, and Figure 5: the structural estimates point to rational reluctance, not inferior ability

Read:

- Table 4 gives the baseline dynamic estimates. The incumbent sunk cost estimate (1.24) is much lower than the entrant sunk cost estimate (2.25), while the fixed operating cost is about 0.15 billion dollars.

	ASSUMED ORDER OF MOVES		
	Old-Both-New-PE (1)	PE-New-Both-Old (2)	PE-Old-Both-New (3)
Fixed cost of operation (ϕ)	.1474	.1472	.1451
	[-.02, .33]	[-.02, .33]	[-.03, .33]
Incumbents' sunk cost (c^{inc})	1.2439	1.2370	1.2468
	[.51, 2.11]	[.50, 2.10]	[.51, 2.11]
Entrants' sunk cost (c^{ent})	2.2338	2.2724	2.2911
	[1.74, 2.85]	[1.76, 2.87]	[1.78, 2.89]
Log likelihood	-112.80	-112.97	-113.46

NOTE.—Figures are in billions of dollars. The 90 percent confidence intervals are in brackets. These intervals are based on likelihood ratio tests (i.e., by comparing $LL(\hat{\theta})$ with the corresponding critical values from the χ^2 distribution, where $LL(\hat{\theta})$ is the joint log likelihood, in P/N, X, L), evaluated at a vector of parameter values $\hat{\theta} = (\hat{\phi}, \hat{c}^{inc}, \hat{c}^{ent})$. $\hat{\theta}$ and $\hat{\theta}$ represent the MLE and its perturbed counterpart, respectively, and represent the lower bounds because they do not incorporate the error from the estimation of static parameters. Standard errors based on Wald-type tests are either implausibly small or large because the likelihood function exhibits a step function-like shape due to the game-theoretic nature of the model.

	$\beta = .75$	$\beta = .85$	$\beta = .90$
Fixed cost of operation (ϕ)	-.04	.35	.55
Incumbents' sunk cost (c^{inc})	1.23	1.39	1.88
Entrants' sunk cost (c^{ent})	2.25	2.50	3.06
Log likelihood	-110.22	-116.84	-127.21
	$\delta = 1.00$	$\delta = 1.05$	$\delta = 1.15$
Fixed cost of operation (ϕ)	.12	.14	.07
Incumbents' sunk cost (c^{inc})	1.59	1.47	1.22
Entrants' sunk cost (c^{ent})	4.26	3.12	1.99
Log likelihood	-124.09	-116.00	-113.60
	$N^{inc} = 1$	$N^{inc} = 2$	$N^{inc} = 3$
Fixed cost of operation (ϕ)	.14	.10	.14
Incumbents' sunk cost (c^{inc})	1.28	1.29	1.28
Entrants' sunk cost (c^{ent})	-.06	1.83	2.09
Log likelihood	-96.94	-103.03	-108.61

NOTE.—As a reminder, the baseline parameter values for β , δ , and N^{inc} are .8, 1.1, and 4, respectively, and the corresponding log likelihood is -112.80.

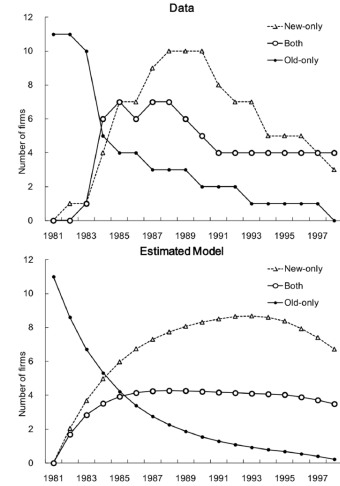


FIG. 5.—Fit of market structure dynamics. The bottom panel displays the mean numbers of firms across 10,000 simulations of the estimated model.

- Alternative order-of-move assumptions barely change the magnitudes.
- Table 5 shows that the ranking $k^{inc} < k^{ent}$ is robust to alternative discount factors, trends in innovation cost, and the assumed number of potential entrants.
- Figure 5 shows that the model fits the broad evolution of old-only, both, and new-only firms fairly well, even if the simulated path is smoother than the data.

Economic message. This is where the paper overturns the naive story. Incumbents do not appear to innovate slowly because they face higher innovation costs. They innovate slowly despite having lower estimated innovation costs.

5.5 Figures 6 and 7: cannibalization dominates preemption in explaining the innovator's dilemma

Read:

- In the no-cannibalization counterfactual, incumbents innovate much faster. Over the first 10 years, incumbent innovations rise from 6.30 in the baseline to 8.95.
- Measured as the cumulative difference between entrant and incumbent innovators by 1998, the innovation gap shrinks from 6.45 firms to 2.80 firms, a reduction of about 57%.
- In the no-preemption counterfactual, incumbents innovate slightly less: only 6.02 incumbents innovate in the first 10 years, and the eventual innovation gap widens by about 38% to 8.91 firms.
- Figure 7 pushes the point further: to match entrants' innovation pace, incumbents would need almost zero sunk cost of innovation.

Economic message. This is the paper's headline decomposition. Preemption matters and pushes in the opposite direction, but cannibalization is quantitatively much more important.

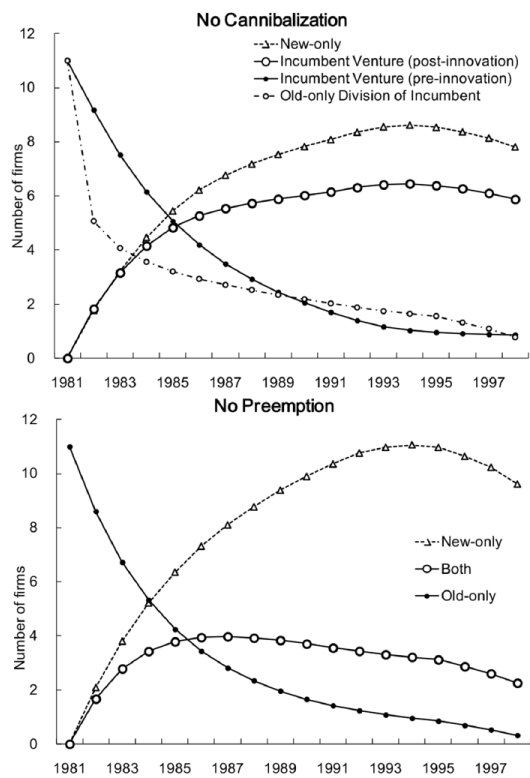


FIG. 6.—Explaining the innovator's dilemma (1). The mean numbers of firms across 10,000 counterfactual simulations.

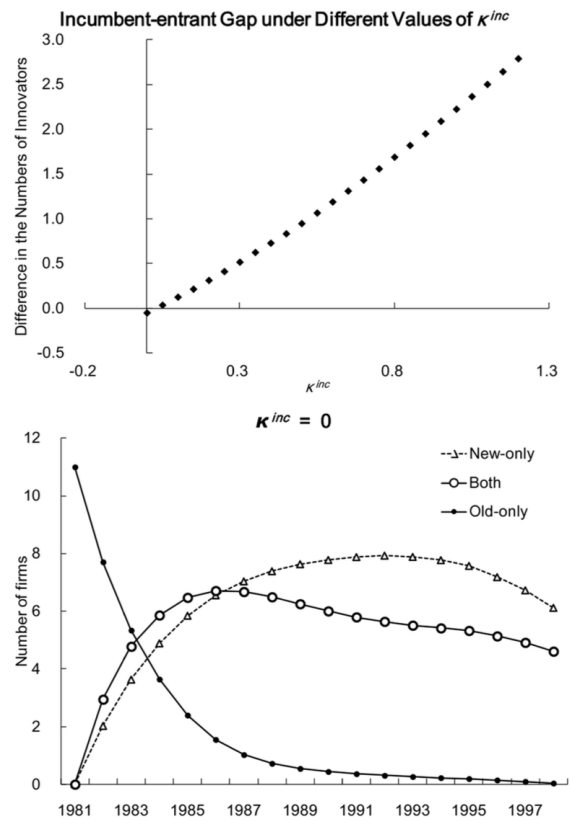


FIG. 7.—Explaining the innovator's dilemma (2). The innovation gap in the top panel refers to the difference in the numbers of innovators among incumbents and entrants during the first 10 years, and a positive gap indicates entrants' lead over incumbents (see eq. [12]). The bottom panel shows the mean numbers of firms across 10,000 counterfactual simulations.

5.6 Table 6, Figure 8, Figure 9, and Table 7: patent policy can help ex ante and hurt badly ex post

TABLE 6
BRIEF HISTORY OF RODIME'S 3.5-INCH HDD PATENT AFFAIR

Year	Events
1980	Rodime became independent from Burroughs's 5.25-inch HDD plant in Glenrothes, Scotland.
1983	Rodime became the first maker to achieve volume production of 3.5-inch HDDs.
1986*	Rodime surprised the industry by obtaining a patent on the concept of a 3.5-inch drive.
	Rodime sued Miniscribe and Conner Peripheral for patent infringement. IBM sued Rodime, which countersued IBM.
1988*	The 3.5-inch patent affair headed for a long tour of the US federal court system.
	Miniscribe opted out by taking a license from Rodime.
1989	Rodime moved to Singapore for production efficiency but neared bankruptcy and obtained some financing. The company completely overhauled top management in early 1989.
1991	Patent affair ended when IBM and Conner Peripheral, as well as Fujitsu and Alps Electric, took licenses. Several other firms were in negotiation. Rodime pursued joint ventures with Japan's JVC and firms in Taiwan and Korea, but in mid-1991 it announced it would file for bankruptcy and cease manufacturing operations.
	It planned to remain active in pursuing licensing revenues from 3.5-inch HDD patents.
1994	High legal expenses and falling license revenues put financial pressure on Rodime.
1995	In September 1995, a US appeals court ruled some of Rodime's patent claims invalid, a ruling in favor of Quantum. Rodime still argued other patent claims were valid, in a separate legal action against Seagate.
1996	Appeals court rulings in 1995 and 1996 appear to have weakened Rodime's negotiating position, but it continues to argue other patent claims are still valid.

SOURCE.—DISK/TREND Reports.

* Key events that motivate my counterfactual simulations. See Igami and Subrahmanyam (2015) for a study of statistical relationships between patents and innovations in the HDD industry.

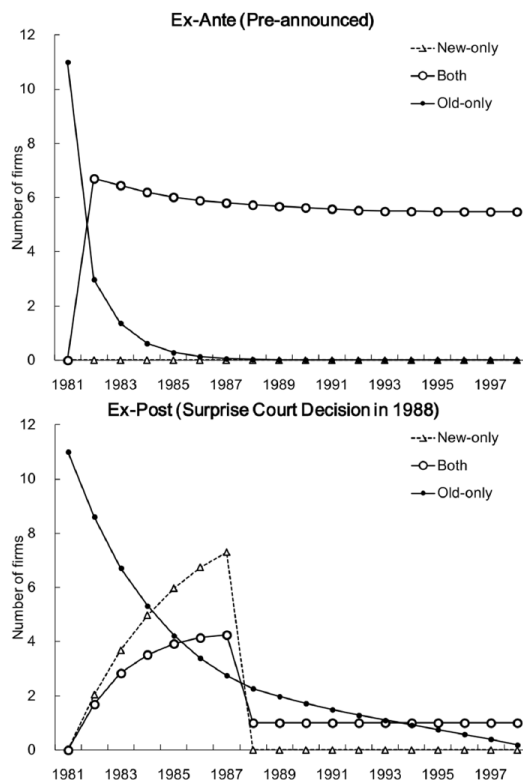


FIG. 8.—Policy experiment 1 (broad patent): the mean numbers of firms across 10,000 counterfactual simulations.

Read:

- Table 6 summarizes the Rodime 3.5-inch patent episode, which motivates the policy counterfactuals.
- Figure 8 compares two broad-patent scenarios. In the ex ante regime, early innovators are protected from the start; in the ex post regime, monopoly rights arrive only in 1988 after rivals have already sunk innovation costs.
- Table 7 shows the welfare arithmetic. Under the ex ante broad patent, social welfare rises from 21.3 to 34.6, an increase of +62.5%. Under the ex post broad patent, welfare collapses from 21.3 to 2.0, a change of −90.8%.
- The license-fee cases are much more nuanced: a 25% license fee raises welfare slightly to 21.5, a 50% fee raises it to 22.0, and a 75% fee lowers it to 20.1.

Economic message. The policy lesson is subtle. Strong ex ante property rights can be good in the model because they reduce duplicated innovation costs. But ex post monopoly grants are disastrous once the real resources have already been spent. That is why the paper treats the Rodime episode as a warning about dysfunctional patent enforcement, not as a simple argument for stronger patents.

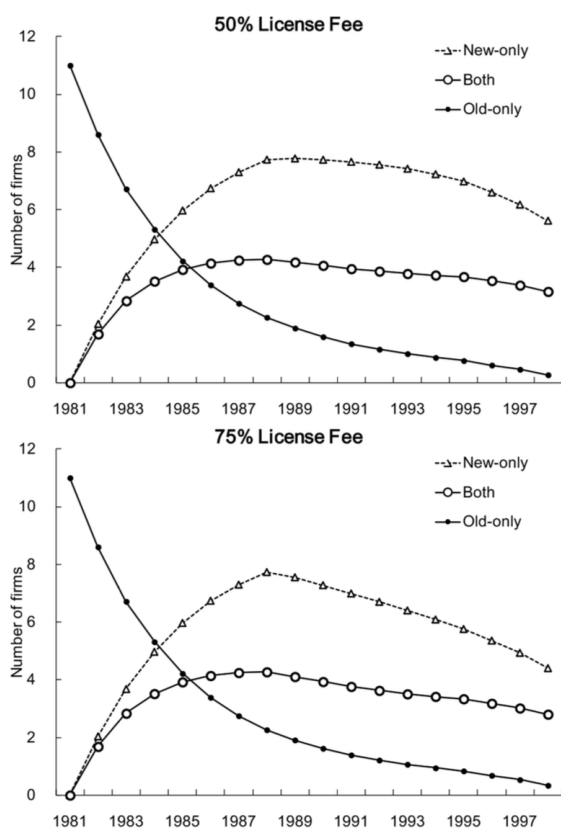


FIG. 9.—Policy experiment 2 (license fees): the mean numbers of firms across 10,000 counterfactual simulations.

TABLE 7
WELFARE ANALYSIS OF INNOVATION/COMPETITION POLICIES

	Consumer Surplus (1)	Producer Surplus (2)	Fixed Cost of Operation (3)	Sunk Cost of Innovation (4)	Social Welfare (1 + 2 + 3 + 4)	Change from Baseline
Baseline	52.7	4.0	-9.7	-25.7	21.3	...
Broad patent:						
Ex ante	42.1	6.9	-5.7	-8.7	34.6	+62.5%
Ex post	28.3	4.9	-8.0	-23.2	2.0	-90.8%
License fees:						
25%	51.2	4.8	-9.6	-25.0	21.5	+7%
50%	50.5	5.2	-9.6	-24.1	22.0	+3.3%
75%	47.2	6.5	-9.4	-24.1	20.1	-5.8%

NOTE.—Each number is the sum of discounted present values as of 1981.

6 Short summary

- The paper studies why incumbents in the HDD industry innovated more slowly than entrants during the shift from 5.25-inch to 3.5-inch drives.
- It estimates a structural model with logit demand, Cournot competition, and a dynamic game of entry, exit, and innovation.
- The preferred estimates imply that incumbents had *lower* sunk innovation costs than entrants, so the observed gap is not driven by inferior innovative ability.
- Cannibalization is the main force behind incumbent reluctance. Removing it shrinks the innovation gap by about 57%.
- Preemption works in the opposite direction and encourages incumbents to innovate, but not enough to dominate the replacement effect.
- The model fits the broad market-structure evolution of the industry and allows the paper to compare policy counterfactuals.
- Ex ante broad patent protection can improve welfare in the model, but ex post monopoly grants can be extremely harmful once rival firms have already sunk innovation costs.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that the innovator's dilemma can arise in a fully rational model of dynamic oligopoly. Incumbents in the HDD industry were slower to innovate than entrants not because they were systematically worse innovators, but because their own old products were profitable enough that new products cannibalized existing rents. The structural estimates and the counterfactual decompositions both point to this conclusion.

7.2 Economic intuition

Arrow's replacement effect is the central intuition. An entrant only sees the upside of the new technology. An incumbent sees both the upside of the new technology and the downside of losing profits on the old technology. Preemption partially offsets this by making early innovation strategically useful, but in the HDD data the cannibalization force is stronger. The result is delayed incumbent innovation even though incumbents enjoy a cost advantage in developing the new technology.

7.3 Takeaway

The deepest takeaway is that slow incumbent innovation need not be evidence of irrationality or organizational failure. It may instead be the efficient response of a profit-maximizing incumbent facing cannibalization. That insight matters not only for business history but also for policy:

interventions that try to stimulate innovation must pay attention to the difference between ex ante incentives and ex post market power.